Habitual dietary sodium intake is inversely associated with coronary flow reserve in middle-aged male twins

Background: Evidence links dietary sodium to hypertension and cardiovascular disease (CVD), but investigation of its influence on cardiovascular function is limited. Objective: We examined the relation between habitual dietary sodium and coronary flow reserve (CFR), which is a measure of overall coronary vasodilator capacity and microvascular function. We hypothesized that increased sodium consumption is associated with lower CFR. Design: Habitual daily sodium intake for the previous 12 mo was measured in 286 male middle-aged twins (133 monozygotic and dizygotic pairs and 20 unpaired twins) by using the Willett food-frequency questionnaire. CFR was measured by positron emission tomography [N13]-ammonia, with quantitation of myocardial blood flow at rest and after adenosine stress. Mixed-effects regression analysis was used to assess the association between dietary sodium and CFR. Results: An increase in dietary sodium of 1000 mg/d was associated with a 10.0% lower CFR (95% CI: −17.0%, −2.5%) after adjustment for demographic, lifestyle, nutritional, and CVD risk factors ($P = 0.01$). Across quintiles of sodium consumption, dietary sodium was inversely associated with CFR (P-trend = 0.03), with the top quintile (>1456 mg/d) having a 20% lower CFR than the bottom quintile (<732 mg /d). This association also persisted within pairs: a 1000-mg/d difference in dietary sodium between brothers was associated with a 10.3% difference in CFR after adjustment for potential confounders ($P = 0.02$). Conclusions: Habitual dietary sodium is inversely associated with CFR independent of CVD risk factors and shared familial and genetic factors. Our study suggests a potential novel mechanism for the adverse effects of dietary sodium on the cardiovascular system. This trial was registered at clinicaltrials.gov as NCT00017836.